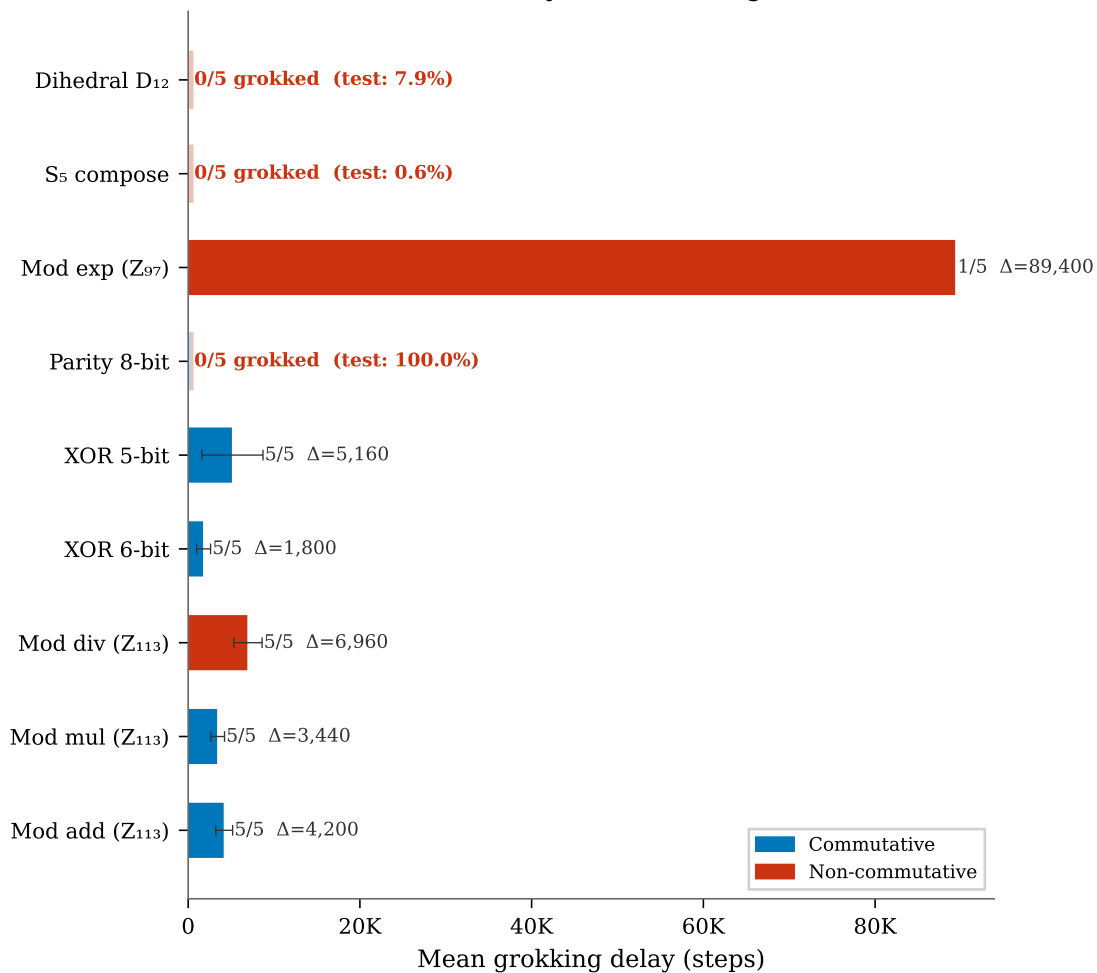


Task taxonomy and per-task circuit analysis (n = 5 seeds)

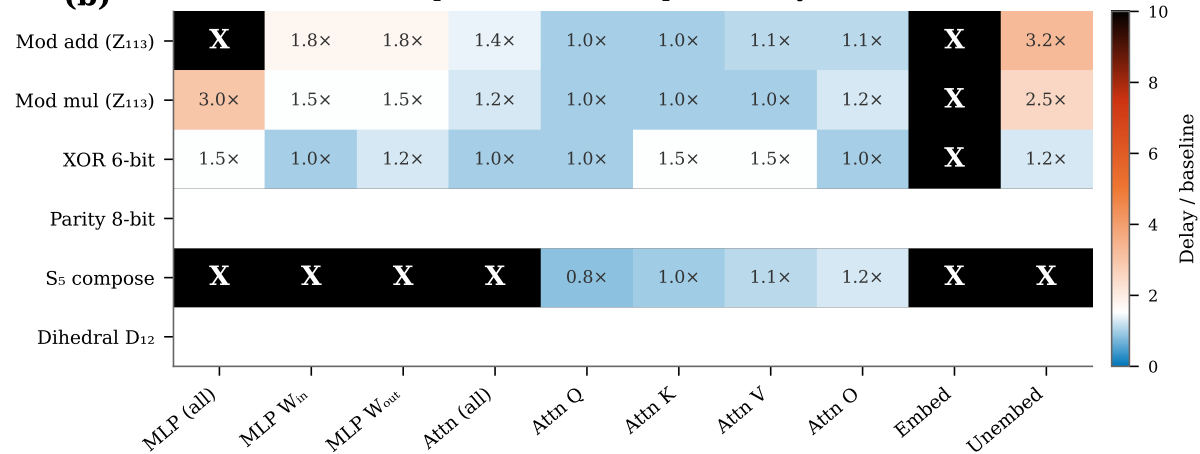
(a)

Task taxonomy: which tasks grok?



(b)

Component freeze impact (delay ratio)



(c)

MLP $\leftrightarrow$ Attention role reversal

